Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/01/2020
PI Name:	Boerma, Marjan Ph.D.		
Project Title:	Effects of Chronic and Single-Dose Charged Particle Exposure on Cardiac Function and Structure in Multiple Animal Models		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	72205-7101	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	08/07/2017	**End Date:**	01/06/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/6/2020 per NSSC info (Ed., 11/12/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Weil, Michael Ph.D. (Colorado State University)		
Grant/Contract No.:	80NSSC17K0425		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Recent evidence suggests that exposure to high-linear energy transfer (LET) radiation encountered during deep space travel may be associated with short- or long-term risks of cardiovascular disease. However, these risks are not well defined. Previous studies with single doses of high-LET radiation in mouse models have shown small but significant changes in cardiac function and structure. While these studies have started to characterize the cardiovascular effects of low-dose high-LET radiation, due to practical considerations the radiation in these experiments had to be administered in a very high dose rate compared to the chronic low-dose rate radiation exposures that astronauts will encounter in deep space. Therefore, additional work needs to be done in animals that are chronically exposed to low-dose rate high-LET radiation. Moreover, the inclusion of additional animal species will greatly enhance the potential for translation of the results to humans compared to research in mouse models alone. Here, we aim to provide insight into the effects of chronic low-dose rate high-LET radiation exposures on the heart and identify how animal species differ in their response to high-LET radiation. We pursue the following aims: 1) In collaboration with the NASA Specialized Center of Research (NSCOR) on Carcinogenesis, we make use of their long-term studies in male and female mouse models exposed to chronic low-dose rate neutrons to assess cardiac function with echocardiography, and we will collect cardiac tissues samples from these mice to determine connective tissue content, infiltration of inflammatory cells, and mitochondrial function; 2) We perform echocardiography and tissue analysis in male rats that are exposed to low-dose rate neutrons in the neutron facility that is part of the NSCOR; 3) We perform echocardiography and tissue analysis in mice exposed to neutrons and silicon ions and treated with aspirin in their drinking water, to determine whether aspirin can protect against radiation-induced changes in the heart; and 4) We assess protein markers of inflammatory cells, of mitochondria and connective tissue structure in hearts of male rats and rabbits previously exposed to protons and oxygen ions and available in our laboratory. These projects make use of existing animals and tissue samples from NASA-funded laboratories to aid in defining the risk of cardiovascular disease from space radiation exposure, and to determine whether aspirin is a potential countermeasure against cardiovascular disease risk from space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There is concern about the potential adverse cardiovascular effects of chronic exposures to low-dose ionizing radiation, such as from medical treatments, occupational low-dose exposures, and radiological accidents. The current project is one of the first to measure cardiac function and structure in animal models of chronic low-dose rate exposures to ionizing radiation. This information will inform not only the risk of radiation exposure during deep-space travel, but also the risks of cardiovascular disease from exposure to low-dose rate radiation exposures on Earth.
Task Progress:	This is the final progress report of this project. All radiation exposures of mice, rats, and rabbits have been completed. In year 2 of the project, echocardiography performed on female BALB/C mice ~400 days after initiating chronic exposures to neutrons had revealed a significant decrease in left ventricular fractional shortening and increase in left ventricular posterior wall thickness in animals that received the highest dose (0.4 Gy). These mice were reassessed 800 days after initiation of the chronic neutron exposure (400 days after completion of the irradiation). At this time point, the changes in left ventricular fractional shortening did not persist. We cannot exclude that this is in part the result of a loss of animals between days 400 and 800, leading to a selection of the healthier mice still alive at day 800. Soon after echocardiography, cardiac tissue was obtained and used to measure mitochondrial function. We observed a trend towards lower mitochondrial oxidative phosphorylation (OXPHOS) in the irradiated groups, together with a trend for higher release of reactive oxygen species (ROS, hydrogen peroxide). Taken together, these data suggest that chronic neutron exposure leads to mild cardiac mitochondrial dysfunction, perhaps through deficient energy production and/or promotion of oxidative stress. In year 2 of the project, echocardiography was performed on male C3H mice ~400 days after initiating chronic exposures to neutrons. Results showed a significant impairment in left ventricular fractional shortening and increase in left ventricular wall thickness, consistent with the neutron-irradiated female BALB/C mice described above. The male C3H mice underwent another cardiac ultrasound at day 800 after initiation of neutron exposure (400 days after completion of the irradiation). Results show that left ventricular posterior wall thickness remained greater in the highest dose (0.4 Gy) group. In addition, left ventricular inner diameter in diastole decreased in this group, which resulted in an increase in fractional shortening. Soon after echocardiography, cardiac tissue was obtained and used to measure mitochondrial function. A significant decrease in OXPHOS was observed after irradiation, which may have contributed to the changes observed in cardiac function. Male C3H mice were exposed to a single dose of silicon ions of 0.4 Gy, and aspirin was administered in the drinking water for 400 days. At 400 days after completion of aspirin treatment, echocardiography and respirometry on cardiac tissue samples were performed. Exposure to silicon ions caused a small decrease in fractional shortening, which coincided with a decrease in cardiac mitochondrial OXPHOS. Both the radiation-induced change in cardiac systolic function and mitochondrial respiration were mitigated by aspirin treatment. Specific Aim 4 was completed in years 1 and 2 of the project. Consistent with mouse and rat studies under Specific Aims 1-3, low-dose charged particle irradiation (protons, 250 MeV, oxygen ions, 600 MeV/n) caused small changes in cardiac contractile function and in mitochondrial morphology, as assessed with electron microscopy in male New Zealand White rabbits. Altogether, these studies suggest that chronic and acute low doses of space-relevant radiation cause only mild changes in long-term cardiac function, and that chronic low-dose aspirin administration may mitigate these effects.
Bibliography Type:	Description: (Last Updated: 01/08/2026)
Articles in Peer-reviewed Journals	Sridharan V, Seawright JW, Landes RD, Cao M, Singh P, Davis CM, Mao XW, Singh SP, Zhang X, Nelson GA, Boerma M. "Effects of single-dose protons or oxygen ions on function and structure of the cardiovascular system in male Long Evans rats." Life Sci Space Res (Amst). 2020 Aug;26:62-8. https://doi.org/10.1016/j.lssr.2020.04.002 ; PMID: 32718688; PMCID: PMC7387753 , Aug-2020

Articles in Peer-reviewed Journals	Upadhyay M, Rajagopal M, Gill K, Li Y, Bansal S, Sridharan V, Tyburski JB, Boerma M, Cheema AK. "Identification of plasma lipidome changes associated with low dose space-type radiation exposure in a murine model." Metabolites. 2020 Jun 17;10(6):E252. https://doi.org/10.3390/metabo10060252 ; PMID: 32560360; PMCID: PMC7345467 , Jun-2020
Articles in Peer-reviewed Journals	Nemec-Bakk AS, Sridharan V, Seawright JW, Nelson GA, Cao M, Singh P, Cheema AK, Singh B, Li Y, Koturbash I, Miousse IR, Ewing LE, Skinner CM, Landes RD, Lowery JD, Mao X-W, Singh SP, Boerma M. "Effects of proton and oxygen ion irradiation on cardiovascular function and structure in a rabbit model." Life Sci Space Res. 2023 May 1;37:78-87. https://doi.org/10.1016/j.lssr.2023.03.008 ; PMID: 37087182; PMCID: PMC10122719 , May-2023